American Computer Science League

2021-2022 ● Contest 4: Fibonacci & Mandelbrot ● Junior Division

PROBLEM: Given the function $f(z) = z^2 + C$, "iterate" the function 5 times. That is, evaluate $f(f(f(f(f(0)))))$, stopping if the absolute value (as defined below) is greater than 4. The variable z and the constant C are *complex* numbers.

In order to use *complex* numbers, we need to define an *imaginary* number, *i*, such that $i = \sqrt{-1}$, and therefore, the value of $i^2 = \sqrt{-1} * \sqrt{-1} = -1$. A *complex* number has two parts: a *real* part and an *imaginary* part. They are represented in the form $a + bi$, where a is referred to as the *real* number part and bi as the *imaginary* number part. Both *a* and *b* are *real* numbers.

There are three operations you will need to perform on *complex* numbers: addition, multiplication, and absolute value. Here is how each operation is performed:

- **Addition.** Combine the *real* number parts and combine the *imaginary* number parts:

  $$(a + bi) + (c + di) = (a + c) + (b + d)i$$

  Example: $(5 + 2i) + (3 - 6i) = 8 - 4i$

- **Multiplication.** Use the distributive property on binomial terms:

  $$(a + bi)(c + di) = ac + adi + bci + bdi^2$$
  $$= (ac - bd) + (ad + bc)i$$

  Example: $(5 + i)(3 + 2i) = 5 \cdot 3 + 5 \cdot 2i + 1 \cdot 3i + 1 \cdot 2i^2$
  $$= 15 + 10i + 3i - 2$$
  $$= 13 + 13i$$

  Example: $(5 + 2i)(3 - 6i) = 27 - 24i$

- **Absolute Value.** The absolute value of a *complex* number is given by the formula:

  $$|a + bi| = \sqrt{a^2 + b^2}$$

  Example: $|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

In this problem, you will be given two decimal numbers separated by a single space. They will represent the real part, *a*, and the imaginary part, *b*. This defines the constant C for the function $f(z) = z^2 + C$. For each input, evaluate up to 5 iterations of the function *f*. While performing the computation on *f*, keep the result of the function in full precision. For each iteration, if the absolute value of the result is greater than 4, stop the process and output "ESCAPES *n*", where *n* is the number of the last iteration that was completed. There is a single

space after the word ESCAPES. Otherwise, round the final absolute value to 3 decimal places. Output that number as a string in the format "x.xxx".

EXAMPLE #1:

Iteration for f(z) = z^2 + (0.3 - 0.67i)	*Absolute Value*
$f(0) = (0+0i)^2 + (0.3 - 0.67i)$ $= 0.3 - 0.67i$	0.73409808608…
$f(f(0)) = f(0.3 - 0.67i)$ $= (0.3 - 0.67i)^2 + (0.3 - 0.67i)$ $= (-0.3589 - 0.402i) + (0.3 - 0.67i)$ $= -0.0589 - 1.072i$	1.07361688231…
$f(f(f(0))) = f(-0.0589 - 1.072i)$ $= (-0.0589 - 1.072i)^2 + (0.3 - 0.67i)$ $= (-1.1457147 + 0.1262816i) + (0.3 - 0.67i)$ $= -0.84571479 - 0.5437184i$	1.0054169306…
$f(f(f(f(0)))) = f(-0.84571479 - 0.5437184i)$ $= (-0.84571479 + 0.5437184i)^2 + (0.3 - 0.67i)$ $= (0.41960387 + 0.91966138i) + (0.3 - 0.67i)$ $= 0.71960387 + 0.24966138i$	0.76168264187…
$f(f(f(f(f(0))))) = f(0.71960387+0.24966138i)$ $= (0.71960387+0.24966138i)^2 + (0.3 - 0.67i)$ $= (0.45549883-0.310685433i) + (0.3 - 0.67i)$ $= 0.75549883-0.310685433i$	0.81688672693…

Because the absolute value of the function is never greater than 4, the output would be **`0.817`**.

EXAMPLE #2:

Iteration for f(z) = z^2 + $(2 - 0.3i)$	*Absolute Value*
$f(0) = (0 + 0i)^2 + (2 - 0.3i)$ $= 2 - 0.3i$	2.02237484...
$f(f(0)) = f(2 - 0.3i)$ $= (2 - 0.3i)^2 + (2 - 0.3i)$ $= (3.91 - 1.2i) + (2 - 0.3i)$ $= (5.91 - 1.5i)$	6.09738468...

Because the absolute value of the function is greater than 4, the evaluation stops and the output would be **`ESCAPES 2`**.

INPUT: Two decimal numbers representing the real part, *a*, and the imaginary part, *b,* of a *complex* number separated by a single space. This defines the constant, C, in the function $f(z) = z^2 + C$.

OUTPUT: For each input, output a string in the format "x.xxx" or "ESCAPES *n*".

SAMPLE INPUT:

```
0.3 -0.67
2.0 -0.3
-0.5 0.56
-0.62 0.43
0.0 0.543
```

SAMPLE OUTPUT:

```
1. 0.817
2. ESCAPES 2
3. 0.018
4. 0.508
5. 0.574
```

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TEST DATA

TEST INPUT:

```
0.45 -0.56
-0.325 0.765
0.561 -0.456
-0.213 0.729
-0.053 0.0
```

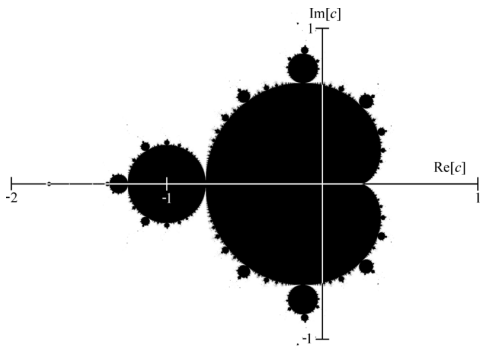
TEST OUTPUT:

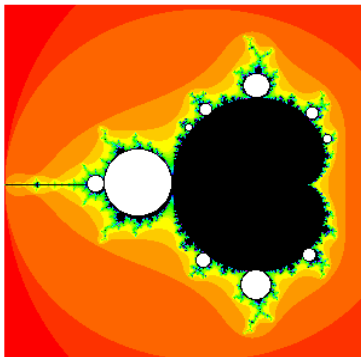
1. 2.153
2. 1.057
3. ESCAPES 5
4. 0.834
5. 0.050

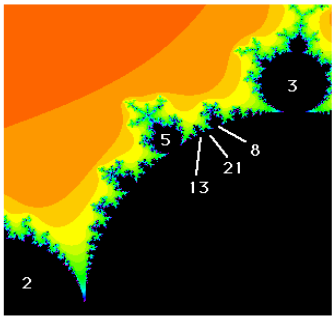
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BACKGROUND INFORMATION FOR THE ADVISOR:







The Mandelbrot Set is generated by iterating the function $f(z) = z^2 + C$ for points within a small range on the complex plane where the point (a,b) represents the complex number a + bi as shown in the 1st illustration.

When iterating a function over itself, the function either converges to a single value, cycles between multiple values, or "escapes" to infinity, meaning that its value gets larger each time. If you are given an initial value for C, a complex number, starting with a value of z = 0, the function generates a sequence of numbers until one of these occurs.

If the iteration results in converging or cycling among 2 or more values, the point is colored back because it is in the Mandelbrot Set. The various colors outside the Mandelbrot Set are used to illustrate how many iterations it takes to exceed some value, thus how quickly the values "escape" to infinity.

The main "bulb" (black area in the 2nd illustration) in the Mandelbrot Set represents all points that converge or have a cycle of length 1. The smaller bulbs (white areas in the 2nd illustration) represent points with different cycle lengths.

The Fibonacci sequence can be found in the Mandelbrot Set as the length of the cycle for all of the points in each of the bulbs in the upper left and lower left sections of the main bulb as shown in the 3rd illustration given.

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问题: 给定**函数** $f(z) = z^2 + C$, 将**该函数**迭代 5 **次**。**也就是计算** $f(f(f(f(f(0)))))$，如果绝对值（如下述定义）大于 4，则停止计算。变量 z 和常量 C 是复数。

为了使用复数，我们需要定义一个虚数, i , 为此，我们定义 $i = \sqrt{-1}$ 。于是可以**得到** $i^2 = \sqrt{-1} * \sqrt{-1} = -1$ ，一个复数有两个部分：实部和虚部，以 $a + bi$ 的形式表示，其中 a 表示实部，bi 表示虚部。a 和 b 都为实数。

你需要对复数执行三种运算：加法、乘法和取绝对值。以下是每种运算的执行方式：

- **加法**. 实部和实部相加，虚部和虚部相加：

  $$(a + bi) + (c + di) = (a + c) + (b + d)i$$

  例如: $(5 + 2i) + (3 - 6i) = 8 - 4i$

- **乘法**. 使用二项式乘法分配律：

  $$(a + bi)(c + di) = ac + adi + bci + bdi^2$$
  $$= (ac - bd) + (ad + bc)i$$

  例如: $(5 + i)(3 + 2i) = 5 \cdot 3 + 5 \cdot 2i + 1 \cdot 3i + 1 \cdot 2i^2$
  $$= 15 + 10i + 3i - 2$$
  $$= 13 + 13i$$

  例如: $(5 + 2i)(3 - 6i) = 27 - 24i$

- **绝对值.** 根据以下公式得出一个复数的绝对值：

  $$|a + bi| = \sqrt{a^2 + b^2}$$

  例如: $|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

在这个问题中，你将得到两个十进制数，它们之间用一个空格隔开。分别表示实部 a 和虚部 b。对于函数 $f(z) = z^2 + C$，C 是常数。 对于每个输入，计算函数 f 的 5 次迭代。在 f 上执行计算时，确保函数的结果准确无误。对于每次迭代，如果得到的结果的绝对值大于 4，则停止计算并输出 "ESCAPES n"，其中 n 是完成的最后一次迭代的次数。单词 ESCAPES 后有一个空格。

否则，将最终得到的绝对值四舍五入到小数点后 3 位。且将该数字以字符串 "x.xxx" 的格式输出。

例子 #1:

迭代 $f(z) = z^2 + (0.3 - 0.67i)$	绝对值
$f(0) = (0+0i)^2 + (0.3 - 0.67i)$ $= 0.3 - 0.67i$	0.73409808608…
$f(f(0)) = f(0.3 - 0.67i)$ $= (0.3 - 0.67i)^2 + (0.3 - 0.67i)$ $= (-0.3589 - 0.402i) + (0.3 - 0.67i)$ $= -0.0589 - 1.072i$	1.07361688231…
$f(f(f(0))) = f(-0.0589 - 1.072i)$ $= (-0.0589 - 1.072i)^2 + (0.3 - 0.67i)$ $= (-1.1457147 + 0.1262816i) + (0.3 - 0.67i)$ $= -0.84571479 - 0.5437184i$	1.0054169306…
$f(f(f(f(0)))) = f(-0.84571479 - 0.5437184i)$ $= (-0.84571479 + 0.5437184i)^2 + (0.3 - 0.67i)$ $= (0.41960387 + 0.91966138i) + (0.3 - 0.67i)$ $= 0.71960387 + 0.24966138i$	0.76168264187…
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因为函数的绝对值都不大于 4，所以输出将为 `0.817`。

例子 #2:

迭代 $f(z) = z^2 + (2 - 0.3i)$	绝对值
$f(0) = (0 + 0i)^2 + (2 - 0.3i)$ $= 2 - 0.3i$	2.02237484...
$f(f(0)) = f(2 - 0.3i)$ $= (2 - 0.3i)^2 + (2 - 0.3i)$ $= (3.91 - 1.2i) + (2 - 0.3i)$ $= (5.91 - 1.5i)$	6.09738468...

因为函数的绝对值大于 4，所以计算停止，输出为 **ESCAPES 2**。

输入：两个十进制数，分别代表复数的实部 a 和虚部 b，它们之间用一个空格隔开。在函数$f(z) = z^2 + C$ 中，定义常数为 C。

输出: 对于每一个输入，输出“ x.xxx ” 或者 “ ESCAPES n ” 格式的字符串。

样本输入:

```
0.3 -0.67
2.0 -0.3
-0.5 0.56
-0.62 0.43
0.0 0.543
```

样本输出:

```
1. 0.817
2. ESCAPES 2
3. 0.018
4. 0.508
5. 0.574
```

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测试数据

测试输入:

```
0.45 -0.56
-0.325 0.765
0.561 -0.456
-0.213 0.729
-0.053 0.0
```

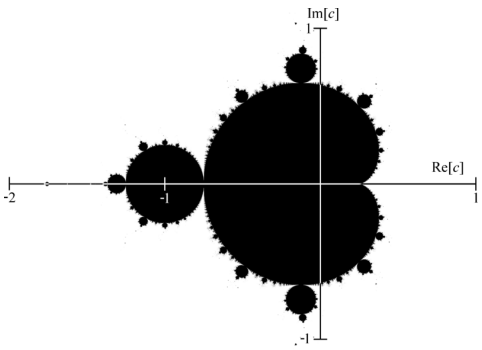
测试输出:

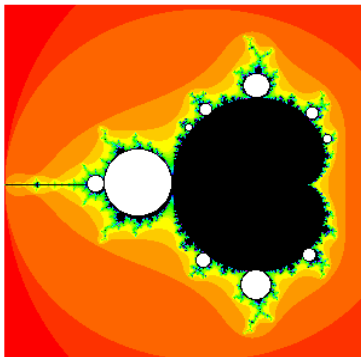
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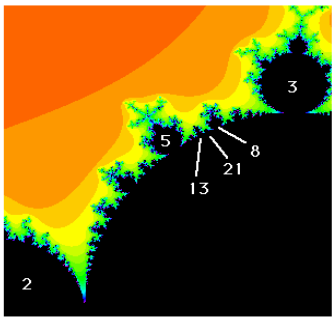
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背景信息:







曼德布洛特**集合是通过迭代函数** $f(z) = z^2 + C$**，**在复**平面上**输出**小**范围内**的点**而生成的，**其中点（**a , b**）表示复数** $a + bi$**，如图 1 所示。**

当函数进行**自**我**迭代时，函数要么收敛**至**一个值，在多个值之间**“ 反复**循环**”，要么**“ escapes ”**到无穷大，这意味着它的值每次都会变大。如果给你一个复数** C **的初始值，**且**从** $z = 0$ **开始，**该**函数会生成一**系列**数字，直到**出现上述现象中的**一个。**

如果迭代的结果是在 2 个或多个值之间收敛或循环，则该点的背景会被着色，因为它位于曼德布洛特**集合中。曼德**布洛特**集合外的各种颜色是用**来**说明需要多少次迭代才能超过某个值，从而说明这个值“ escapes ”到无穷大的速度**有多快。

曼德布洛特**集合中的“ 主**灯泡区域**”（第二幅图中的黑色区域）表示所有收敛或循环长度为 1 的点。较小的“ 灯泡区域”（第二幅图中的白色区域）代表具有不同循环长度的点。**

如给出的第 3 幅图所示，在“ 主灯泡区域 ”的左上部分和左下部分中，每个“ 灯泡区域 ”中表示循环长度的所有点都能够在曼德布洛特集合中找到斐波那契数列。